Chapter 5 Binomial Theorem, Sequences and Series

"Life stands before me like an eternal spring with new and brilliant clothes"

Johann Carl Friedrich Gauss

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5.1 Introduction

Binomial theorem facilitates the algebraic expansion of the binomial $(a + b)$ for a positive integral exponent n. Binomial theorem is used in all branches of Mathematics and also in other Sciences. For example using the Binomial theorem one can easily find the coefficient of x^{20} in the expansion of $(2x - 7)^{23}$. If one wants to know the maturity amount after 10 years on a sum of money deposited in a nationalised bank at the rate of 8% compound interest per year or to know the size of population of our country after 15 years if the annual growth rate and present population size are known, Binomial theorem helps us in finding the above quantities. The coefficients in the binomial expansion of $(a + b)^n$, $n \in \mathbb{N}$, are called binomial coefficients. Binomial theorem plays a vital role in determining the probabilities of events when the random experiment involves finite sample space and each outcome is either success or failure. In this chapter we learn binomial theorem and some of its applications.

Greek Mathematician Euclid mentioned the special case of binomial theorem for exponent 2. Binomial theorem for exponent 3 was known by 6th century in India. In 1544, Michael Stifel (German Mathematician) introduced the term **binomial coefficient** and expressed $(1 + x)^n$ in terms of $(1 + x)^{n-1}$.

The German Mathematician Johann Carl Friedrich Gauss is one of the most renowned Mathematicians in history. Many have referred to him as the "Prince of Mathematics ". He has contributed in the areas of Number theory, Physics, Astronomy etc., Number Theory was Gauss's favourite field and he referred to Number theory as the "Queen of Mathematics". Anecdote involves, his school teacher who wanted to test the students asked them to sum the integers from 1 to 100. Within a few seconds Gauss shown the answer has 5050. Nobody is sure which method of summing an arithmetic sequence Gauss used as a child.



Johann Carl Friedrich Gauss (1777–1855)

Over the period of thousand years, legends have developed mathematical problems involving sequences and series. One of the famous legends about series concerns the invention of chess, where the cells of chess board were related to $1, 2, 4, 8, \ldots$ (imagine the number related to 64th cell). There are many applications of arithmetic and geometric progressions to real life situations.

In the earlier classes we have learnt about sequences, series. Roughly speaking a sequence is an arrangement of objects in some order and a series is the sum of the terms of a sequence of numbers. The concept of infinite series helps us to compute many values, like $\sin \frac{9}{44}\pi$, $\log 43$ and e^{20} to a desired level of approximation. Sequences are important in differential equations and analysis. We learn more about sequences and series.